

Multiplication of Positive Integers Practice Worksheet

1. 6×7
2. 8×24
3. 35×16
4. 142×28
5. 325×412
6. $4 \times 15 \times 3$
7. $25 \times 16 \times 4$
8. $8 \times 15 \times 25 \times 4$
9. An auditorium has 38 rows of seats, with each row containing 24 seats. What is the total seating capacity of the auditorium?
10. A warehouse receives a shipment of 125 pallets. Each pallet contains 48 boxes, and each box holds 16 individual items. What is the total number of items in the shipment?

Solutions Key: Multiplication of Positive Integers

1. **Answer:** 42
Explanation: Basic single-digit multiplication fact: $6 \times 7 = 42$.
2. **Answer:** 192
Explanation: Multiply 8 by the digits of 24: $8 \times 4 = 32$ (write 2, carry 3), $8 \times 2 = 16$ plus 3 equals 19, giving 192.
3. **Answer:** 560
Explanation: Use long multiplication: $35 \times 6 = 210$, and $35 \times 10 = 350$. Adding these partial products gives $210 + 350 = 560$.
4. **Answer:** 3,976
Explanation: Multiply 142 by 8 to get 1,136, then multiply 142 by 20 to get 2,840. Summing the partial products: $1,136 + 2,840 = 3,976$.
5. **Answer:** 133,900
Explanation: Multiply 325 by each digit of 412 from right to left: $325 \times 2 = 650$; $325 \times 10 = 3,250$; $325 \times 400 = 130,000$. Summing the results: $650 + 3,250 + 130,000 = 133,900$.
6. **Answer:** 180
Explanation: Use the associative property to multiply efficiently: $4 \times 15 = 60$, then $60 \times 3 = 180$.
7. **Answer:** 1,600
Explanation: Reorder the factors to make computation easier using numbers that multiply to powers of ten: $25 \times 4 = 100$, then $100 \times 16 = 1,600$.
8. **Answer:** 12,000
Explanation: Group factors strategically to simplify the multiplication: $(25 \times 4) = 100$ and $(8 \times 15) = 120$. Then, multiply the resulting values: $120 \times 100 = 12,000$.
9. **Answer:** 912 seats
Explanation: Multiply the number of rows by the number of seats per row: $38 \times 24 = 912$ seats.
10. **Answer:** 96,000 items
Explanation: Multiply the number of pallets, boxes per pallet, and items per box: $125 \times 48 \times 16$. Grouping $125 \times 16 = 2,000$, then multiplying by 48 gives $2,000 \times 48 = 96,000$ items.